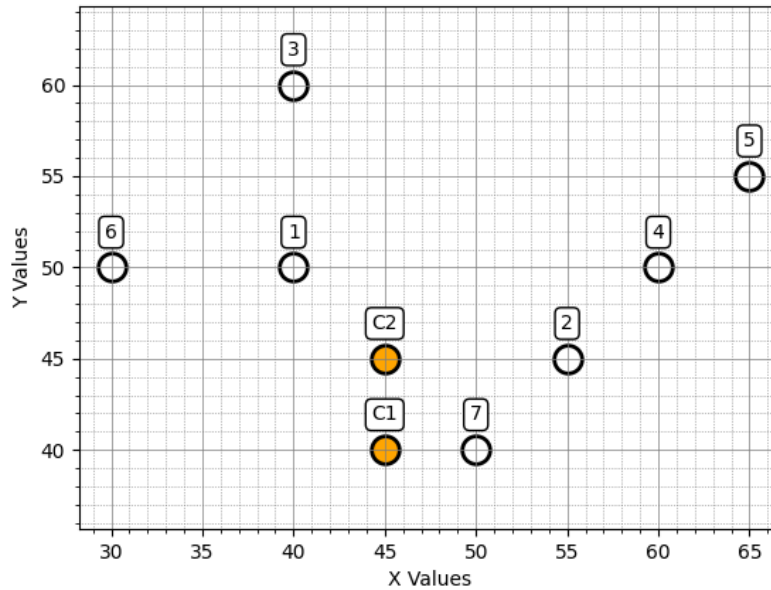


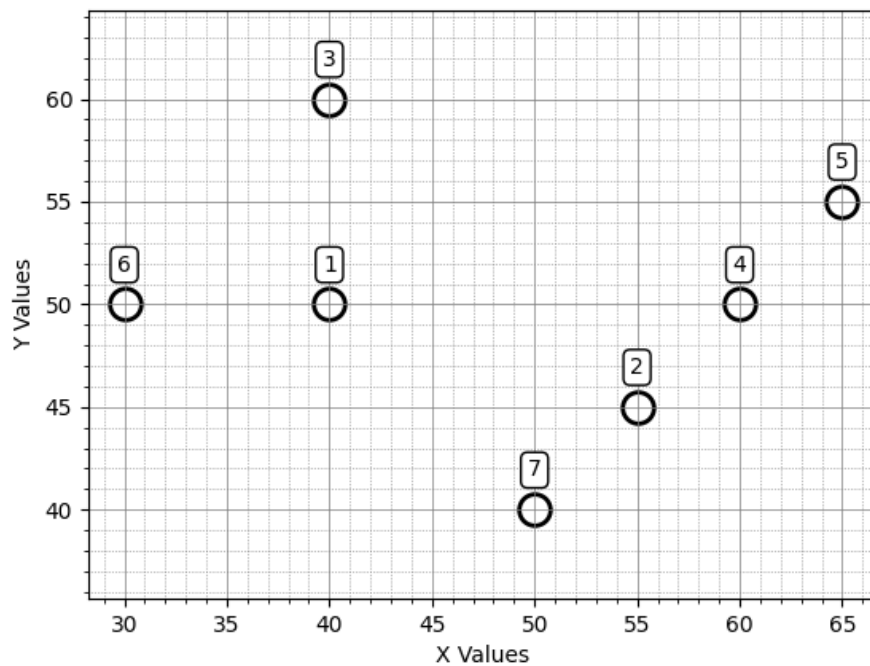
1 K-mean clustering

Apply the k-mean clustering to divide the data points into two clusters, where we have initialized its centroids (highlighted points) as shown in the 1st iteration; then, perform the k-mean clustering to assign the cluster to each data point for **two** iterations. 16 points

1. 1st iteration (**7 scores**):



	Points (x, y)	Cluster #1	Cluster #2	Assigned Cluster (#1 or #2)
		Distance to centroid (C1) (x, y) = (40, 45)	Distance to centroid (C2) (x, y) = (45, 45)	
1.	(40, 50)			
2.	(50, 50)			
3.	(40, 60)			
4.	(60, 50)			
5.	(65, 55)			
6.	(30, 50)			
7.	(50, 40)			

2. 2nd iteration (**9 scores**):

Show how to compute centroid for cluster #1:

Show how to compute centroid for cluster #2:

	Points (x, y)	Cluster #1	Cluster #2	Assigned Cluster (#1 or #2)
		Distance to centroid (x, y) = (...)	Distance to centroid (x, y) = (...)	
1.	(40, 50)			
2.	(50, 50)			
3.	(40, 60)			
4.	(60, 50)			
5.	(65, 55)			
6.	(30, 50)			
7.	(50, 40)			